Contribution Summary

SAMPLE FORMAT – replace all example content with your own. Keep it to 1–2 pages.

Name	Teammate A (example)	Version	v1.0 (first push)
Module	Data preprocessing	Date	DD-MM-YYYY
Folder	/teammate_a/preprocessing/	Depends on	None (first stage of pipeline)

1. What I built and why

• Built a preprocessing module that cleans raw network flow data (CSV) and outputs model-ready features.
• **Why:** the detection agent needs numeric, scaled, class-balanced input; raw flows have NaNs, infinities and mixed types.
• **Why this approach:** StandardScaler chosen over MinMax because flow features have heavy outliers.
• **AI usage:** AI drafted the cleaning functions; I verified the logic and tested on a 10k-row sample.

2. Files pushed

File	Purpose
`loader.py`	Reads CSV, validates column names, returns a DataFrame
`cleaner.py`	Drops duplicates, handles NaN / inf values, fixes dtypes
`features.py`	Selects features, encodes labels, scales values
`main.py`	Runs loader -> cleaner -> features end to end

3. Data flow

`raw.csv -> loader.load() -> cleaner.clean() -> features.build() -> (X_train, X_test, y_train, y_test)`

Function	Input	Output	Key parameters
`load(path)`	str: CSV path	DataFrame, shape (n, 79)	`path`: file location
`clean(df)`	DataFrame (raw)	DataFrame (no NaN/inf)	`drop_dupes=True`
`build(df)`	DataFrame (clean)	4 arrays: X_train, X_test, y_train, y_test	`test_size=0.2`, `seed=42`

4. Key decisions / logic

• Infinity values replaced with column max, NaNs with median (median is robust to outliers).
• Labels converted to binary (0 = benign, 1 = attack) for the first version; multi-class can be added later.
• Scaler is fitted on the training set only, to avoid data leakage.

5. Changes to earlier code (skip in the first push)

What changed	What went wrong	What others must adjust
`build()` now returns 4 arrays instead of 2	Test data was being scaled with its own statistics (leakage)	Update every call to unpack 4 values: `X_train, X_test, y_train, y_test = build(df)`
Label column renamed `Label -> label`	Case mismatch broke the loader on Linux	Use lowercase `label` in any code reading this column

6. How to run / dependencies

```
python main.py --input data/raw.csv --output data/processed/
```

Requires: pandas, numpy, scikit-learn (see `requirements.txt`). Output is saved to `data/processed/`.

7. Known limitations / next steps

• No handling yet for categorical protocol fields (planned for v1.1).
• Tested only on a sample, not the full dataset.